

Summary of the Live Professor Q&A on Regression at the 2026 Astrostatistics Summer School (06/02/2026)

1 Quick recap

This was a Q&A session following a regression tutorial that Eric Feigelson substituted for Ashley Villar, who was unable to attend due to having a baby. Eric provided additional insights beyond Ashley's lectures, emphasizing that many astronomical problems involve nonlinear regression rather than linear regression, and highlighting the importance of including error terms (ϵ) in regression equations. He discussed advanced topics including errors-in-variables models, maximum likelihood estimation, and various regression methods beyond least squares such as L1 regression and LASSO. Students asked questions about measurement errors in both dimensions, hypothesis testing for regression parameters, cross-validation versus correlation functions, and the differences between cross-entropy loss and least squares for logistic versus linear regression. Eric stressed the importance of residual analysis, using multiple statistical methods for reliability, and examining likelihood surfaces through corner plots to ensure parameter uniqueness.

2 Summary

2.1 Astronomy Regression Techniques Lecture

Eric Feigelson substituted for Ashley Villar to discuss regression techniques in astronomy, building on Ashley's previous lectures. He emphasized that while Ashley's content was good, regression in astronomy often involves nonlinear models, heteroscedastic errors, and non-Gaussian distributions, which go beyond basic linear regression. Eric highlighted important concepts including R-squared coefficients, residual analysis, logistic regression, model selection tools like BIC, and non-parametric regression methods like LOESS and Gaussian processes. He noted that there is much more depth to the field than could be covered in a brief lecture.

2.2 Errors in Variables Models Discussion

George raised a question about measurement errors in both dimensions of the response variable, which Eric addressed by explaining that such cases are known as Errors in

Variables Models (EIV) in statistics. Eric mentioned that while simple bivariate cases can be handled with existing methods like ODR (Orthogonal Distance Regression), multivariate analysis with multiple X variables becomes more complex. Eric also noted that some of this content is covered in their textbook "Modern Statistical Methods" which he co-authored.

2.3 Regression Parameter Value Determination

Hrishabh asked about determining the true value and confidence intervals for regression parameters, to which Eric explained that the true value is never known and confidence intervals can be challenging to determine accurately using different methods. Stephen inquired about installing Astrolib R, which Eric clarified was already addressed in the Hash tutorials Slack chat by TA Das. Eric encouraged students to ask future questions in the Slack channel for broader discussion and support.

2.4 Model Validation Methods Discussion

Eric explained the difference between heuristic models and astrophysical models, noting that while both can have more parameters, overfitting must be avoided using methods like the Bayesian Information Criterion (BIC). He clarified that cross-validation emerged from bootstrap concepts in the 1970s-80s as a way to estimate confidence intervals and reliability of regression parameters, particularly useful in machine learning applications. Eric mentioned that while the textbook covers bootstrap adequately, cross-validation could be improved in the second edition.

2.5 Multiple Statistical Methods in Astronomy

Eric discussed the importance of using multiple statistical methods when analyzing data in astronomy, emphasizing that practitioners should run different approaches like least squares, LASSO, and others to ensure scientific results are reliable. He recommended using R or Python to easily implement multiple methods and create histograms of parameter distributions from bootstrap or cross-validation procedures to assess the robustness of findings. Eric noted that while statisticians may disagree on approaches, combining non-parametric methods like bootstrap with parametric procedures can provide effective solutions for data analysis challenges.

2.6 Cross Entropy in Regression Models

Prabhanjan asked about the application of cross entropy in logistic regression versus linear regression, which Eric couldn't fully answer but suggested trying different objective functions to see their impact on results. Hyungsuk explained that cross entropy loss functions are suitable for classification tasks due to their connection with distributional assumptions like Bernoulli or multinomial distributions, while least squares is used for regression assuming Gaussian errors. Eric also emphasized the importance of residual analysis, particularly testing for normality, to ensure the appropriateness of using least squares regression. Yash inquired about the uniqueness of solutions when fitting high-order polynomials, though the discussion ended without a complete answer.

2.7 Multimodal Distributions in Statistical Analysis

Eric discussed the challenges of multimodal likelihood distributions in statistical analysis, explaining that while unimodal distributions provide reliable results, multimodal distributions can lead to unreliable scientific judgments. He suggested using corner plots from the EMCEE code to identify multimodal issues and recommended that when faced with multiple modes, scientists should consider discussing alternative physical interpretations of their results. Eric concluded by announcing that he would be returning later to discuss supervised and unsupervised learning.